SEMINAR III · WEEK 3

Numbers That Make Sense

This week's big question: **does my answer make sense?**

Numbers Tell a Story

A positive or negative sign is not just decoration. It tells you something real about a situation.

- **Money:** +$50 gained, −$50 owed
- **Temperature:** 8°F above zero, −8°F below
- **Elevation:** +100 ft above sea level, −100 ft below
- **Change:** +12 increase, −12 decrease

Before you calculate anything, ask what the sign actually means in the situation.

Which Number Is Greater?

On a number line, numbers farther to the right are greater. $-8 < -2 < 0 < 3 < 10$. Every positive number is greater than every negative number. Zero is neither positive nor negative.

> **Common mistake:** assuming a negative number with a bigger digit is bigger. It's the opposite — −12 is less than −5, not greater. Think about debt: would you rather owe $5 or owe $12? A balance of −$5 is better than −$12.

Absolute value is the distance a number is from zero. $|-8| = 8$ and $|8| = 8$, because both are 8 units from zero. It is never negative, and it does not mean "make the number positive." A temperature of −4°F does not become 4°F just because its absolute value is 4 — it means −4°F is four degrees away from zero.

Working With Positive and Negative Numbers

Addition, same direction: $-5 + (-3)$. Start at −5, move 3 more negative. Result: −8.

Addition, opposite directions: $-5 + 8$. Start at −5, move 8 positive. Result: 3.

Subtraction: 8 − (−3) = 11. Subtracting a negative removes a negative quantity. If a $3 loss is removed, your position improves by $3.

Same signs → positive result

Different signs → negative result

This shortcut applies to multiplication and division only — it does not transfer to addition and subtraction.

Which Operation Do I Need?

Addition: combined, total, gained, increased by

Subtraction: difference, decreased, remaining

Multiplication: equal groups, rate × quantity

Division: split equally, amount per group

Keywords are clues, not guarantees. Always ask: what quantities do I have, what am I trying to find, and how are the quantities related?

What Happens First? (Order of Operations)

P — Parentheses (or other grouping symbols)

E — Exponents

M/D — Multiplication and Division, left to right

A/S — Addition and Subtraction, left to right

Example: 6 + 3 × 4. Multiply first: 3×4=12. Then add: 6+12=**18** — not 36.

Common misconception: multiplication always happens before division. It doesn't — equal priority, left to right. 20 ÷ 5 × 2 = 4 × 2 = **8**, not 20÷(5×2)=2. Same for addition/subtraction: 12−5+3 = 7+3 = **10**, not 12−(5+3).

Estimate Before You Trust

Estimating gives you a rough target so you can catch mistakes. Example: estimate 49 × 21 by rounding to 50 × 20 = 1,000. The exact answer should land near 1,000 — if a calculator shows

10,290, that should immediately look wrong. Estimating is not random guessing; it's rounding to numbers you can work with easily, then reasoning from there.

Using a Calculator Well

> Calculators are useful for lengthy arithmetic, decimals, and checking your work. They cannot decide which operation a situation needs, what a question is actually asking, or whether your answer is reasonable. If you meant to calculate 52×18 but typed 52×81, the calculator will correctly answer the wrong question. Estimate first, and check what you entered.

Does My Answer Make Sense?

Check	Ask
Operation	Did I use the correct operation for this situation?
Size	Is the answer approximately the size I expected?
Sign	Should the answer be positive, negative, or zero?
Unit	Did I answer in the correct unit (dollars, degrees, items)?
Context	Does this answer actually make sense in the situation?

Learn From the Error

Not every wrong answer means the same thing. Use the five error types from Week 2:

Knowledge
"I don't remember the rule for multiplying negative numbers." Response: review the concept.

Process
Using the wrong order of operations. Response: review the steps and retry.

Execution
Setting up 42×6 correctly but making an arithmetic mistake. Response: check the calculation.

Comprehension

Calculating the final amount when the question asked for the amount of change. Response: reread the question.

Strategy

Doing a long exact calculation when estimating would have been faster. Response: consider a more efficient approach.

Quick Reference

Number line: negative numbers, then zero, then positive numbers, increasing left to right.

Absolute value: distance from zero.

Multiplication/division signs: same signs → positive, different signs → negative.

Order of operations: Parentheses, Exponents, Multiplication/Division left to right, Addition/Subtraction left to right.

Check your answer: Operation, Size, Sign, Unit, Context.

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